

Pollinator Crop Labeling Guidelines

1. Purpose and Scope

This document defines labeling rules for candidate insect crop images generated from Arctic field images by the pollinator detection preprocessing pipeline. Labels are created using `crop_label.py` and used to train a binary classifier and a group classifier.

2. Labeling Tool

Launch with the following command:

```
python3 crop_labeler.py \  
  --results Insects_images/results_N \  
  --output Insects_images/labeled_crops
```

Progress is saved automatically after every few actions. Quit any time by pressing Q and resume later. Progress per camera folder is saved in `labeled_crops/progress/{cam_name}_result.json`. The last viewed image is saved in `lab_crops/session.json` and restored on the next launch.

Layout of the Tool

Left: full camera frame (debug image) with bounding boxes overlaid. Each box is colour-coded by label.

Click anywhere on the top panel to open a full-screen debug overlay showing the frame at maximum size. In the overlay:

- Click a bounding box to select the corresponding crop
- Press a label key to assign a label
- Press any key to close the overlay

Note: The bottom-left panel contains toggle buttons that control display and navigation behaviour. These options do not change saved labels.

- **Label all:** When ON, label keys apply to all unlabeled crops if no crop is selected
- **Arrows:** Switch Left/Right keys between image navigation and crop navigation
- **After:** Set behaviour after labeling (none / same / next)
- **Tint:** Adjust colour overlay on labeled crops (visual only)
- **Unlabeled only:** Show only crops without labels
- **Skip labeled:** Skip labeled crops when navigating
- **Left labels / nums / boxes / BG boxes:** Toggle bounding box display options in the left panel
- **Left click opens:** Click bbox to open preview
- **2nd crop opens:** Click selected crop again to open preview
- **Filter row:** Show crops by label (BB / Fly / Butterfly / Other / BG / Unsure)
- **?:** Show or hide help panel

Right: crops extracted from the current frame, shown one row at a time.

Each crop displays:

- index number
- pixel dimensions
- filename
- label badge

The page indicator bar at the very bottom shows how many rows of crops exist and which row is currently visible. If the header shows **MORE CROPS BELOW**, scroll down to avoid missing crops.

3. Navigation Modes

The tool has two navigation modes, toggled with R, and a separate Label all mode (toggled with Tab / E or the bottom button).

IMAGE_NAV mode (default, no badge in header):

- Left / Right arrows — previous / next image
- Up / Down arrows — scroll the crop strip one row at a time
- Label key with no crop selected — labels all unlabeled crops in this image only when Label all mode is ON. If Label all mode is OFF, no crop is labelled unless a crop is selected. Pressing the same key again clears all crops already carrying that label.

CROP_NAV mode (header shows "CROP NAV" badge):

- Left / Right arrows — move crop selection left / right
- Up / Down arrows — move crop selection up / down by row
- Label key — label only the selected crop, then automatically advance to the next crop

4. Controls

Key / Action	Effect
Click crop	Select it
Double-click crop	Large preview
Click top panel	Full-screen debug overlay
b	background
1	bumblebee

2	fly
3	butterfly
4	other
u	unsure
R	Toggle CROP NAV mode
C	Clear all labels in current image
P	Large preview of selected crop
Left / Right	Previous / next image (IMAGE_NAV) or prev/next crop (CROP_NAV)
Up / Down	Scroll crop strip (IMAGE_NAV) or move crop selection by row (CROP_NAV)
Mouse wheel	Scroll crop strip
Q	Quit and save

To change a label: re-select the crop and press the new key. If the header shows MORE CROPS BELOW, scroll down — do not miss crops.

5. Labeling Classes

Label	Key	When to use
bumblebee	1	Bumblebee (Bombus) visible
fly	2	Any fly (Diptera) visible
butterfly	3	Butterfly or moth visible
other	4	Any other insect visible (e.g. beetle)
background	b	No insect visible — includes blurry, overexposed, tiny, or unclear crops; vegetation, soil, shadows, reflections, camera info strips, and static objects
unsure	u	Something moving could be an insect but is undecidable even after checking surrounding frames; insect present but class is unclear and needs verification by a domain expert

Use unsure cautiously. When uncertain between a pollinator class and unsure, prefer the pollinator class, missing a real visit is worse than one uncertain crop. Unsure crops are excluded from classifier training.

6. Labeling Rules

Pollinator classes (1 / 2 / 3 / 4):

- Majority body part visible
- Blurry or partially outside the crop boundary, still label as the appropriate pollinator class
- Identity confirmed by surrounding frames even if not clearly visible in this crop

background (b):

- No insect visible — includes blurry, overexposed, tiny, or unclear crops
- Vegetation, soil, rocks, flowers, shadows, reflections

unsure (u):

- Moving object that could be an insect, undecidable even after checking surrounding frames
- Insect present but class is unclear; needs verification by a domain expert

7. Using Surrounding Frames

The pipeline detects movement between consecutive frames, so an insect typically appears in a small number of frames before disappearing. Always check surrounding frames for ambiguous crops.

Press D to move to the next frame — does the same region appear again

Press A to check the previous frame — was the object already there

A region that appears and disappears over 1–5 frames is very likely a real insect visit.

8. Workflow

- Examine the top debug panel to understand which regions were detected.
- For uncertain crops, check surrounding frames using A and D.
- Label each crop using b / 1 / 2 / 3 / 4 / u.
- In IMAGE_NAV mode, enable Label all mode (Tab / E or the bottom button), then press b with no crop selected to quickly label all unlabeled crops as background when the frame clearly contains no insects.
- Check the page indicator bar at the bottom — if more rows exist, scroll down to ensure no crops are missed.
- Press Q to quit and save at any time.
- For questions about specific cases, discuss in the group channel or consult the domain expert.